

Software Requirements Specification

FX Trading Tracker Platform | Version 2.0

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Table of Contents

- 1. Introduction
- 2. System Architecture
- 3. Technology Stack
- 4. Functional Requirements
- 5. API Specification
- 6. Data Models
- 7. Security Requirements
- 8. User Interface Specifications
- 9. Deployment Architecture
- 10. Integration Points

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the technical requirements for the FX Trading Tracker platform, a web-based application for managing foreign exchange deal tickets across trader, treasury, and administrative roles.

1.2 Scope

The system consists of a React single-page application frontend, a FastAPI REST backend, MongoDB/Couchbase database, and Emergent/S3-compatible object storage. It supports the full deal lifecycle with settlement proof management, 8 report types, and comprehensive audit logging.

2. System Architecture

2.1 Architecture Overview

The application follows a three-tier architecture:

- **Presentation Layer** — React 19 SPA with Shadcn/UI components, TailwindCSS, Axios HTTP client
- **Application Layer** — FastAPI REST API with Pydantic validation, JWT authentication, ReportLab PDF generation
- **Data Layer** — Switchable database (MongoDB via Motor / Couchbase Enterprise SDK) with abstraction layer; Switchable object storage (Emergent / S3-compatible via boto3)

2.2 Component Diagram

Component	Technology	Responsibility
Frontend SPA	React 19, TailwindCSS, Shadcn/UI	User interface, form validation, API calls
API Server	FastAPI, Uvicorn	REST endpoints, business logic, JWT auth
Database Abstraction	database.py, db_mongo.py, db_couchbase.py	Switchable DB backend with cursor wrappers
Storage Abstraction	storage.py	Switchable file storage (Emergent/S3)
Report Engine	reports.py, ReportLab	PDF/CSV generation with branded templates
MongoDB	Motor (async driver)	Document storage, text indexes
Couchbase	SDK 4.6, N1QL	Alternative enterprise DB with scopes/collections

3. Technology Stack

3.1 Backend

Package	Version	Purpose
FastAPI	0.110.1	REST API framework
Uvicorn	0.25.0	ASGI server
Pydantic	2.12.5	Data validation
Motor	3.7.1	Async MongoDB driver
Couchbase SDK	4.6.0	Couchbase Enterprise client
boto3	1.42.58	S3-compatible storage
ReportLab	4.4.10	PDF generation
python-jose	3.4.0	JWT token handling
bcrypt	4.3.0	Password hashing
python-docx	1.1.2	DOCX generation
openpyxl	3.1.5	Excel file parsing

3.2 Frontend

Package	Version	Purpose
React	19.0.0	UI framework
React Router	7.5.1	Client-side routing
TailwindCSS	3.4.17	Utility-first CSS
Shadcn/UI	latest	Component library (Radix-based)
Axios	1.8.4	HTTP client
Recharts	2.15.3	Charts and data visualization
date-fns	4.1.0	Date formatting
sonner	2.0.3	Toast notifications

4. Functional Requirements

FR-001: Authentication & Authorization

ID	Requirement
FR-001.1	System shall authenticate users via email/password with JWT tokens
FR-001.2	System shall enforce role-based access (Trader, Treasury, Admin, Sysadmin)
FR-001.3	JWT tokens shall expire after 24 hours
FR-001.4	Passwords shall be hashed with bcrypt

FR-002: Deal Management

ID	Requirement
FR-002.1	Traders shall create deal tickets with: client name, transaction type (Buy/Sell), transfer type, deal/value dates, currency, amount, rate, bank/crypto account
FR-002.2	System shall auto-generate sequential reference numbers (FX-YYYYMMDD-NNNN)
FR-002.3	Deals shall support 6 transfer types with conditional form logic
FR-002.4	Crypto accounts shall require network selection (SOLANA/ETHEREUM/TRON)
FR-002.5	FX-Intercompany deals shall have dual Ours sections (Selling + Buying)
FR-002.6	FX Bank Deal shall hide destination fields
FR-002.7	FX - Corporate Settlement shall behave identically to FX Local

FR-003: Treasury Processing

ID	Requirement
FR-003.1	Treasury shall view pending deals in a queue with customizable columns
FR-003.2	Treasury shall confirm or return deals with mandatory remarks
FR-003.3	Confirmation shall require ≥ 1 settlement proof (client or processor)
FR-003.4	Returned deals shall be editable and resubmittable by the original trader

FR-004: Settlement Proofs

ID	Requirement
FR-004.1	Traders upload client proofs (images/PDF, max 10MB)
FR-004.2	Treasury uploads processor proofs
FR-004.3	Client proofs not applicable for FX Bank Deal
FR-004.4	Proofs stored via storage abstraction (Emergent or S3)

FR-005: Reference Data

ID	Requirement
FR-005.1	Admin CRUD for: FX Clients, Banks, Counterparties, Currencies, Transaction/Transfer Types
FR-005.2	Bank accounts managed per-bank with Account Name (required) + Account Number

ID	Requirement
FR-005.3	Bank account selector shows searchable autocomplete with name + number
FR-005.4	CSV/Excel import for FX Clients (name, code, type) and Bank Accounts
FR-005.5	Global bank account import supporting fld_* legacy format with auto bank creation

FR-006: Reports

ID	Requirement
FR-006.1	8 report types: Deal Blotter, Settlement, Open Positions, Audit Trail, User Activity, Volume Summary, Client Activity, TMS
FR-006.2	Server-side pagination with sort for linear reports
FR-006.3	CSV and branded PDF export
FR-006.4	Admin-configurable role-based report permissions
FR-006.5	FX-Intercompany dual-line handling in Settlement/TMS reports

5. API Specification

5.1 Authentication

Method	Endpoint	Description
POST	/api/auth/register	Create new user (admin only)
POST	/api/auth/login	Authenticate, returns JWT
PUT	/api/auth/change-password	Change current user password

5.2 Deals

Method	Endpoint	Description
POST	/api/deals	Create deal ticket
GET	/api/deals	List deals (filtered by role)
GET	/api/deals/treasury	Treasury pending queue
GET	/api/deals/{id}	Deal detail
PUT	/api/deals/{id}	Edit returned deal
POST	/api/deals/{id}/confirm	Treasury confirm
POST	/api/deals/{id}/return	Treasury return
POST	/api/deals/{id}/cancel	Trader cancel
POST	/api/deals/{id}/resubmit	Trader resubmit
POST	/api/deals/{id}/proof	Upload settlement proof
DELETE	/api/deals/{id}/proof/{pid}	Delete proof

5.3 Reference Data

Method	Endpoint	Description
GET/POST/PUT/DELETE	/api/reference/{type}	CRUD for companies, banks, currencies, etc.
GET/POST	/api/reference/banks/{id}/accounts	List/create bank accounts
POST	/api/reference/banks/{id}/accounts/import	Per-bank CSV/Excel import
POST	/api/reference/bank-accounts/import	Global multi-bank import (fld_* format)
POST	/api/reference/companies/import	FX Client CSV/Excel import

5.4 Reports

Method	Endpoint	Description
GET	/api/reports/{type}?format=json	Report data (paginated JSON)
GET	/api/reports/{type}?format=csv	Full CSV export
GET	/api/reports/{type}?format=pdf	Branded PDF export
GET/PUT	/api/reports/permissions	Report permission config

6. Data Models

6.1 Deal Document

Field	Type	Required	Description
id	UUID	Yes	Unique identifier
reference_number	String	Auto	FX-YYYYMMDD-NNNN format
client_name	String	Yes	Client/company name
transaction_type	String	Yes	Buy or Sell
transfer_type	String	Yes	FX Crypto, FX Local, etc.
counterparty	String	No	PJL Group entity (FX Local/Interco)
buy_currency	String	Yes	Currency code
currency_amount	Float	Yes	Principal amount
rate	Float	Yes	Exchange rate
amount	Float	Yes	Converted amount (currency_amount x rate)
from_type	String	Yes	bank or crypto
from_network	String	Cond.	SOLANA/ETHEREUM/TRON (if crypto)
to_network	String	Cond.	Destination network (if crypto)
ours_network	String	Cond.	Ours network (if crypto)
status	String	Yes	pending/confirmed/returned/cancelled
settlement_proofs	Array	No	List of proof objects

6.2 Bank Account Document

Field	Type	Description
id	UUID	Unique identifier
bank_id	UUID	Foreign key to bank
account_number	String	Bank account number
account_name	String	Descriptive name (required)
currency_code	String	PHP, USD, etc. (from import)
account_type	String	SA, CA, etc. (from import)
bank_address	String	Branch address (from import)
account_alias	String	Short alias (from import)

7. Security Requirements

ID	Requirement
SEC-01	All API endpoints require JWT authentication except /auth/login
SEC-02	Passwords hashed with bcrypt (work factor 12)
SEC-03	Role-based access enforced on every endpoint
SEC-04	File uploads validated for type and size (10MB max)
SEC-05	CORS configured for known frontend origin
SEC-06	All state-changing actions logged in audit trail
SEC-07	Environment variables used for all secrets (no hardcoded credentials)
SEC-08	Database credentials never exposed to frontend

8. User Interface Specifications

8.1 Login Page

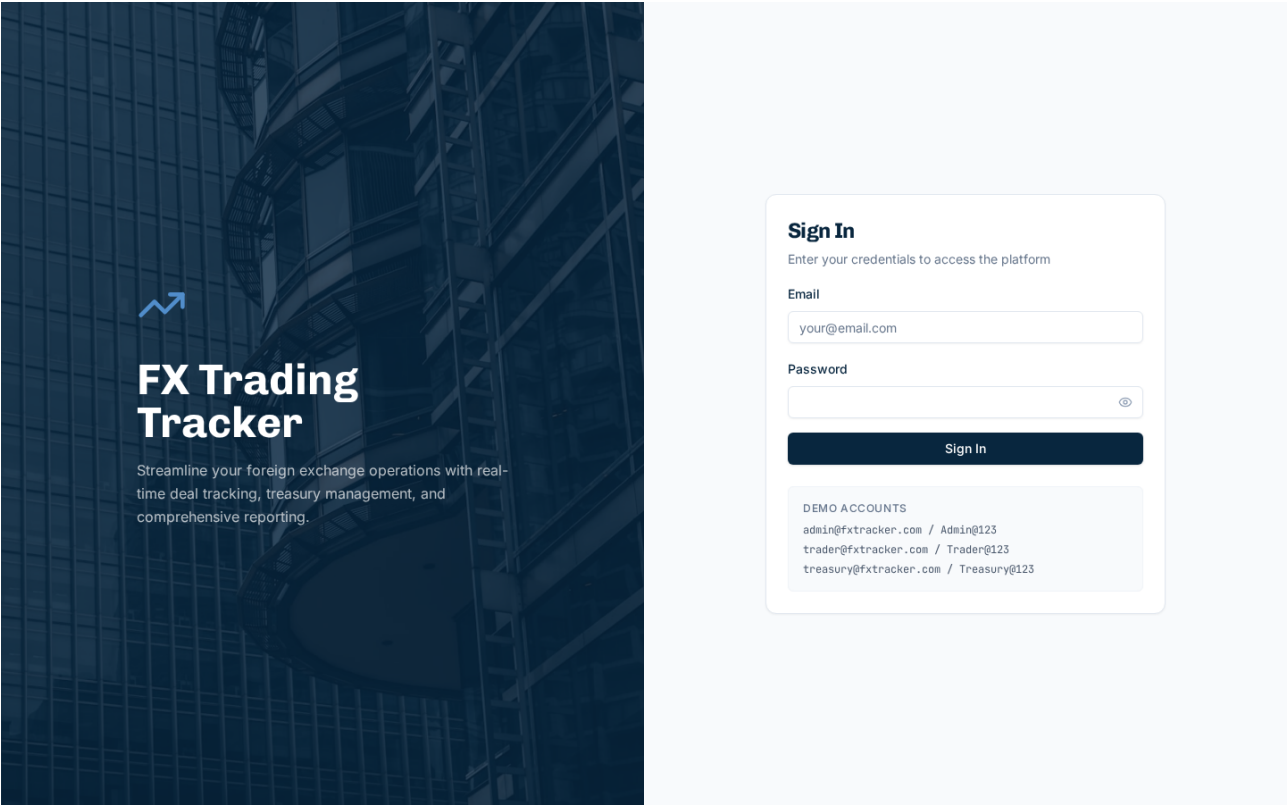


Figure 8.1 - Login Page

8.2 Dashboard

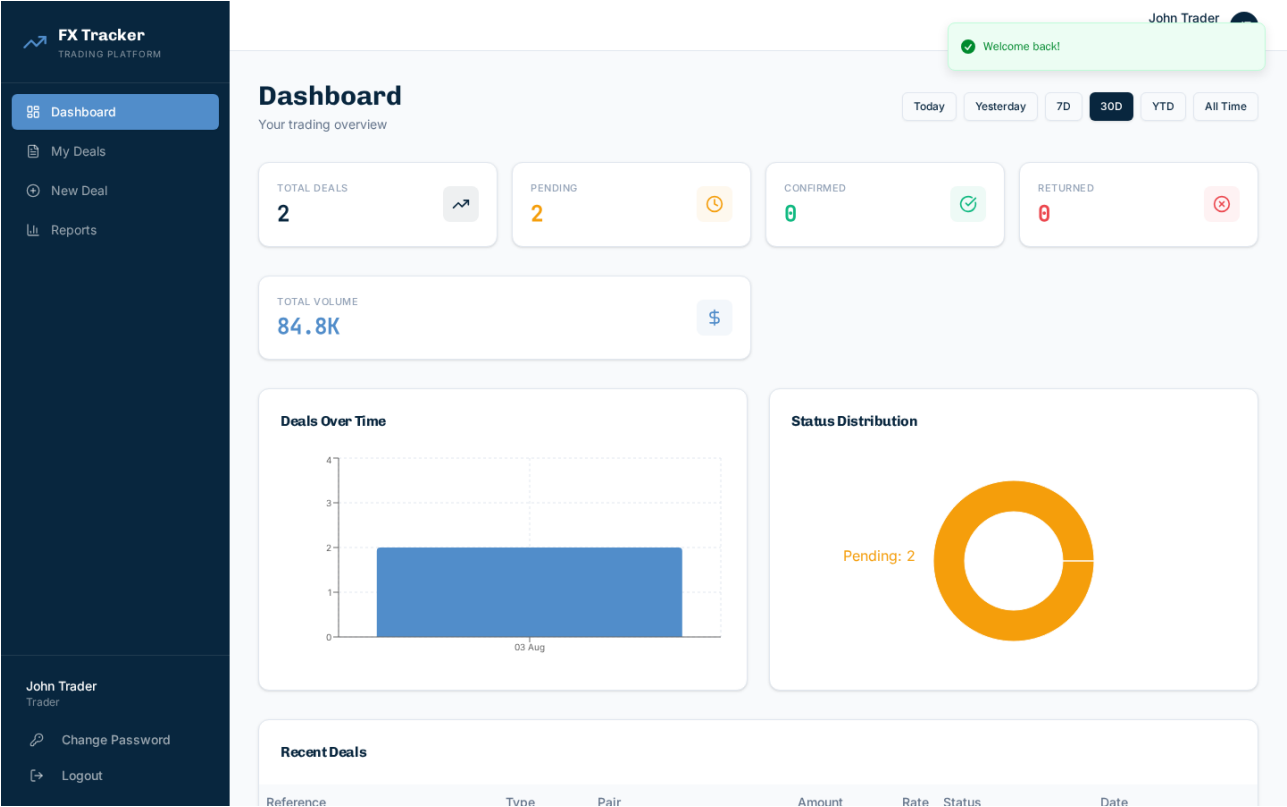


Figure 8.2 - Dashboard with summary cards and charts

8.3 Deal Creation

FX Tracker

TRADING PLATFORM

Dashboard

My Deals

New Deal

Reports

John Trader

Trader

Change Password

Logout

New Deal Ticket

Create a new FX trade deal

Deal Information

Client Name

Enter client name

Transaction Type

Select...

Transfer Type

Select...

Deal Date

03 Aug 2026

Value Date

03 Aug 2026

Amounts & Currency

Currency

Search currency...

Currency Amount

0.00

Exchange Rate

0.000000

Converted Amount

0.00

Source (From)

BankCrypto

Company

Search company...

Bank

Search bank...

Account

Select a bank first

Destination (To)

BankCrypto

Company

Search company...

Bank

Search bank...

Account

Select a bank first

Figure 8.3 - New Deal Form (top)

FX Tracker

TRADING PLATFORM

Dashboard

My Deals

New Deal

Reports

John Trader

Trader

Change Password

Logout

New Deal Ticket

Create a new FX trade deal

Deal Information

Client Name

Enter client name

Transaction Type

Select...

Transfer Type

Select...

Deal Date

24 Aug 2026

Value Date

24 Aug 2026

Amounts & Currency

Currency

Search currency...

Currency Amount

0.00

Exchange Rate

0.000000

Converted Amount

0.00

Source (From)

BankCrypto

Company

Search company...

Bank

Search bank...

Account

Select a bank first

Destination (To)

BankCrypto

Company

Search company...

Bank

Search bank...

Account

Select a bank first

Figure 8.4 - New Deal Form (source/destination)

Page 12

8.4 My Deals

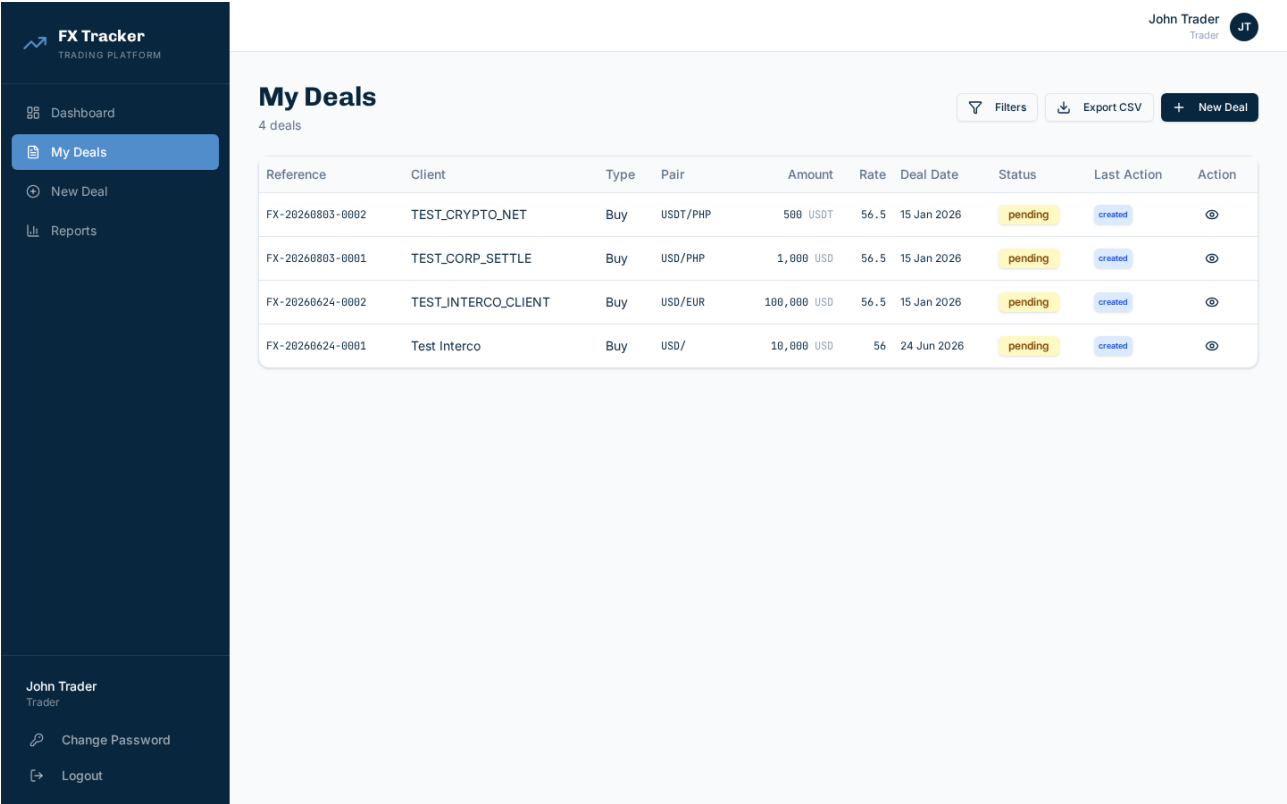


Figure 8.5 - My Deals with currency_amount and Last Action

8.5 Deal Detail

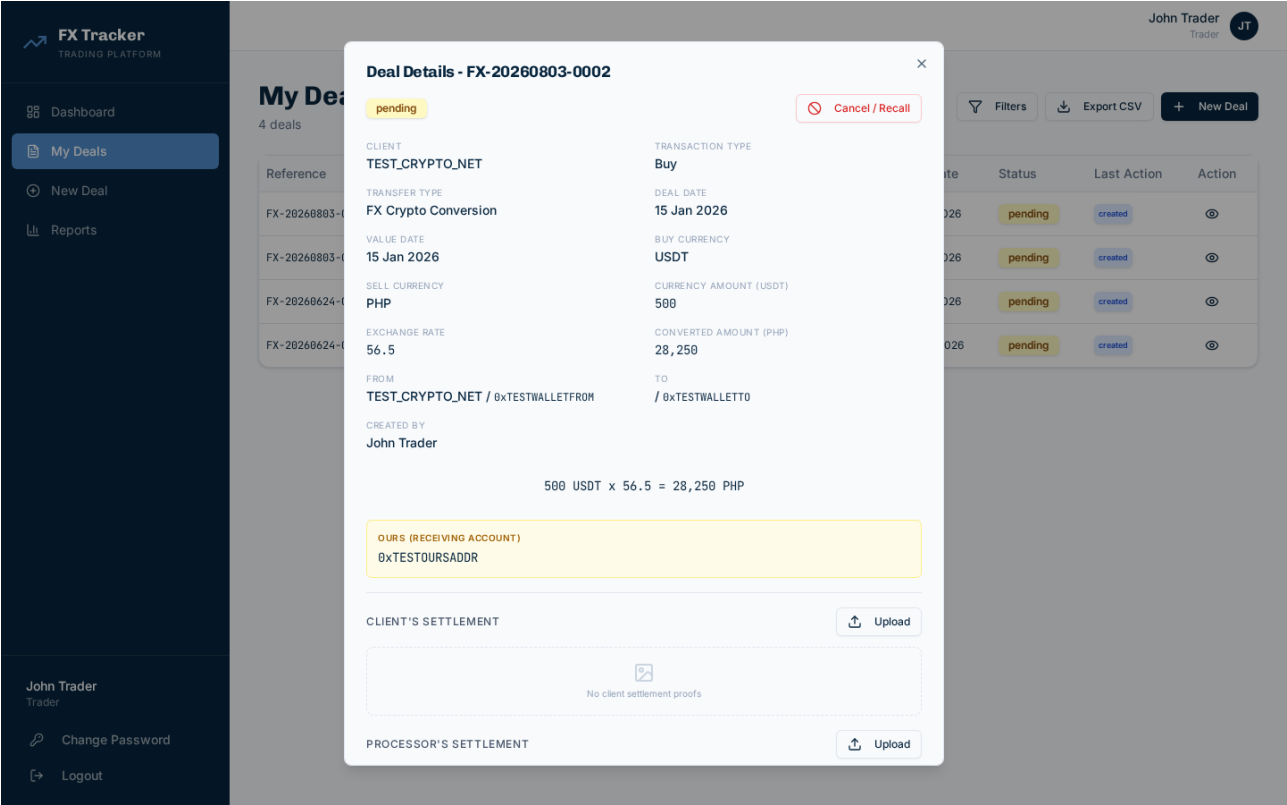


Figure 8.6 - Deal Detail Dialog

8.6 Treasury Queue

FX Tracker
TRADING PLATFORM

Dashboard

Deal Queue

Reports

Jane Treasury
Treasury Ops

Change Password

Logout

Jane Treasury
Treasury Ops

Deal Queue

Review and process FX deal tickets

Pending (4)Returned (0)Processed (0)

Reference	Client	Type	Pair	Amount	Rate	From Bank	To Bank	Deal Date	Last Action	Action
FX-20260803-0002	TEST_CRYPTQ_NET	Buy	USD/PHP	500 USD	56.5	-	-	15 Jan 2026	created	Review
FX-20260803-0001	TEST_CORP_SETTLE	Buy	USD/PHP	1,000 USD	56.5	BDO	BDO	15 Jan 2026	created	Review
FX-20260624-0002	TEST_INTERCO_CLIENT	Buy	USD/EUR	100,000 USD	56.5	JP Morgan Chase	JP Morgan Chase	15 Jan 2026	created	Review
FX-20260624-0001	Test Interco	Buy	USD/	10,000 USD	56	JPMC	Citi	24 Jun 2026	created	Review

Figure 8.7 - Treasury Deal Queue

8.7 Reference Data

FX Tracker
TRADING PLATFORM

Dashboard

Reference Data

Users

Transactions

Audit Trail

Reports

System Admin
Admin

Change Password

Logout

System Admin
Admin

Reference Data

Manage system reference data

Import CSV/ExcelAdd New

FX ClientBanksCounterpartiesTransaction TypesTransfer TypesCurrencies

Name	Code	Status	Actions
Acme Corporation	ACME	Active	
GlobalTech Inc	GTECH	Active	
Pacific Trading Co	PACIFIC	Active	
Sterling Enterprises	STERL	Active	
Atlantic Financial Group	AFG	Active	
Test Company One	TCOMP1	Active	
Test Company Two	TCOMP2	Active	

Figure 8.8 - Reference Data (FX Client)

FX Tracker

TRADING PLATFORM

Dashboard

Reference Data

Users

Transactions

Audit Trail

Reports

System Admin

Admin

Change Password

Logout

System Admin

Admin

SA

Reference Data

Manage system reference data

FX Client

Banks

Counterparties

Transaction Types

Transfer Types

Currencies

Name	Code	SWIFT Code	Status	Actions
JP Morgan Chase	JPMC	CHASUS33	Active	Accounts

| Citibank | CITI | CITIUS33 | Active | Accounts |

| HSBC | HSBC | HSBCGB2L | Active | Accounts |

| Deutsche Bank | DB | DEUTDEFF | Active | Accounts |

+ Add New

Figure 8.9 - Banks with Accounts

8.8 Reports

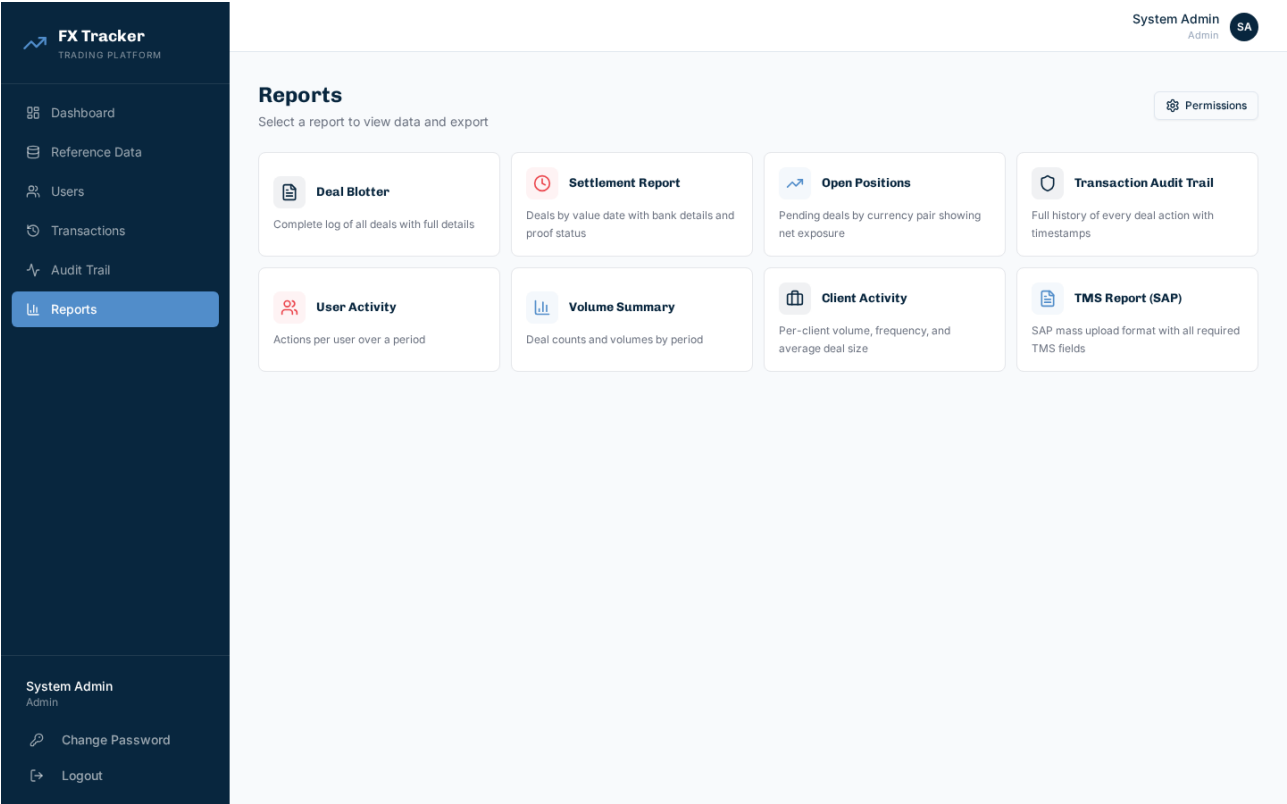


Figure 8.10 - Reports Page

8.9 Audit Trail

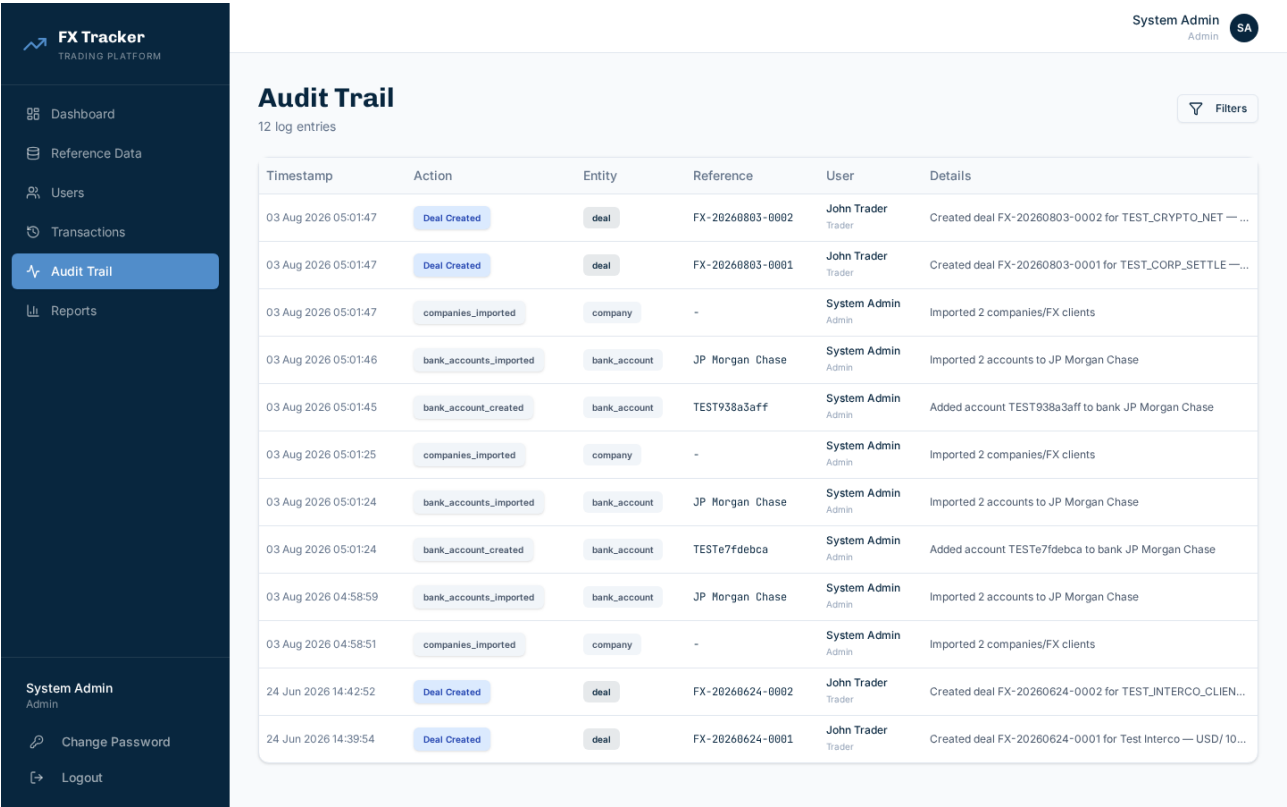


Figure 8.11 - Audit Trail

8.10 System Administration

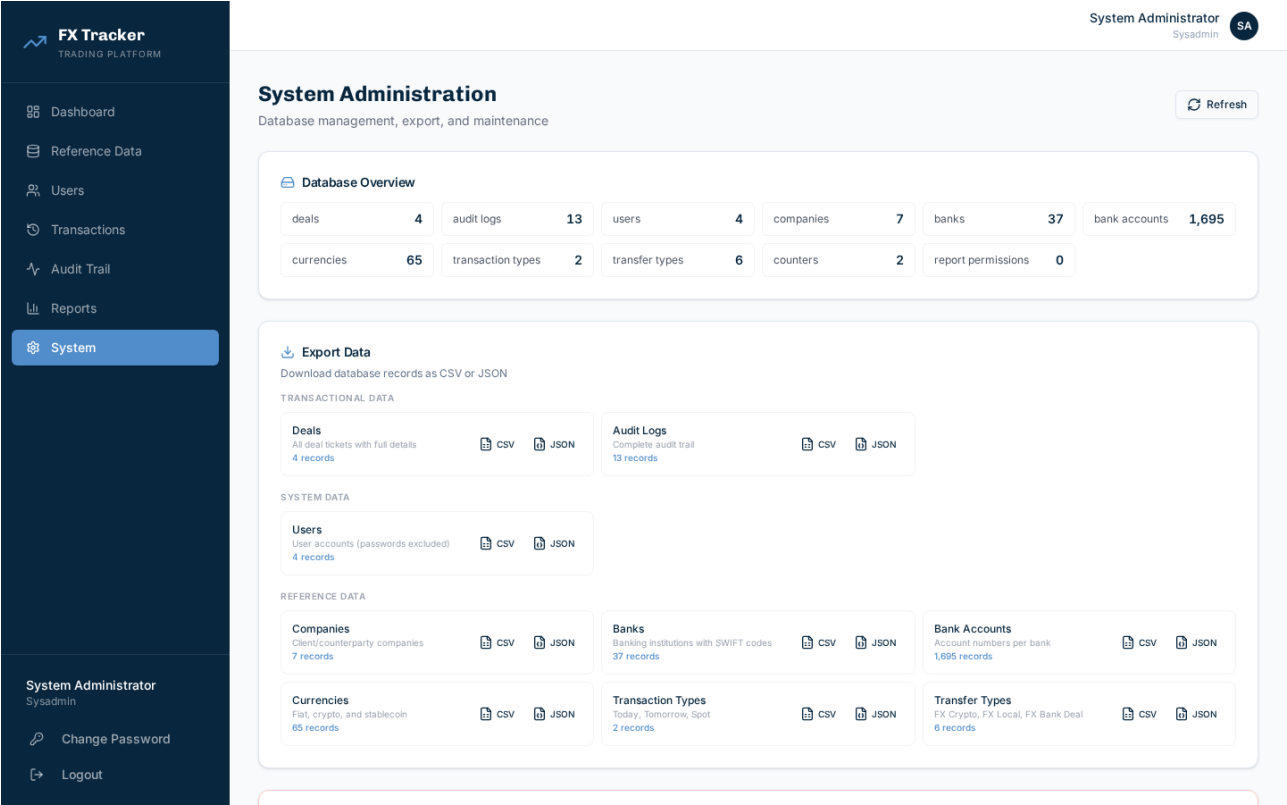


Figure 8.12 - System Admin Page

9. Deployment Architecture

9.1 Environment Configuration

Variable	Location	Purpose
REACT_APP_BACKEND_URL	frontend/.env	API base URL for frontend
MONGO_URL	backend/.env	MongoDB connection string
DB_NAME	backend/.env	Database name
DB_TYPE	backend/.env	mongodb or couchbase
STORAGE_TYPE	backend/.env	emergent or s3
S3_ENDPOINT / S3_ACCESS_KEY / S3_SECRET	backend/.env	S3-compatible storage config
CB_CONNECTION_STRING / CB_USERNAME / etc.	backend/.env	Couchbase connection config

9.2 Process Management

Backend and frontend are managed by Supervisor with automatic hot-reload. Backend runs on port 8001, frontend on port 3000, with Kubernetes ingress routing /api/* to backend.

10. Integration Points

Integration	Protocol	Purpose	Status
MongoDB	Motor (async)	Primary database	Active
Couchbase Enterprise	SDK 4.6 / N1QL	Alternative enterprise DB	Validated
Emergent Object Storage	HTTP API	Settlement proof storage	Active
S3-Compatible (Huawei OBS / MinIO)	MinIO / S3 API	On-premise proof storage	Validated
ReportLab	Python library	PDF report generation	Active
openpyxl	Python library	Excel import parsing	Active